Model-Based vs. Model-Free Reinforcement Learning for Autonomous Driving

Nihal Abdul Naseer, Mohammed Alawadhi, and Abdulhafedh Al-Zubedi

***Abstract*—Reinforcement learning for autonomous driving has largely been dominated by model-free methods, while recent model-based algorithms have reported sample-efficiency advantages on robotic-control benchmarks. Building on our prior survey of model-based and model-free reinforcement learning, this work empirically compares the two paradigms in multi-agent driving and asks whether model-based advantages transfer beyond the domains in which they were originally validated. We deploy TD-MPC2, a recent model-based algorithm from ICLR 2024, on the `highway-env` simulator and compare it against Soft Actor-Critic (SAC) across one training environment, `highway-v0`, and two unseen environments, `merge-v0` and `roundabout-v0`. We hypothesize that learned world models, by encoding environment dynamics rather than scenario-specific policies, should transfer better than they fit. Our experiments provide behavioral evidence consistent with this hypothesis. While TD-MPC2 underperforms SAC on the training environment under our compute budget (0.0% vs. 71.1% success), it retains 85% of its training-environment reward when transferred to unseen scenarios, compared with only 21% for SAC—a $4.0\times$ larger transfer factor. On `roundabout-v0`, TD-MPC2 achieves a 53.3% success rate compared with SAC's 43.3% (Cohen's $d = 0.88$, large effect; every TD-MPC2 seed exceeds the SAC mean). A latent-space dynamics probe further reveals a counter-intuitive separation: the model's state-prediction accuracy does not transfer to unseen environments, even though its planning-derived behavior does. This suggests that the transferable component may be action-level abstraction rather than low-level dynamics fidelity. To our knowledge, this is the first benchmark of TD-MPC2 on multi-agent driving, and the results highlight a decoupling between in-distribution performance and zero-shot generalization. More broadly, we suggest that model-based algorithms should be evaluated using both prediction-level and behavior-level transfer metrics. Code, checkpoints, and analyses are publicly available.**

***Index Terms*—Model-based reinforcement learning, model-free reinforcement learning, autonomous driving, zero-shot transfer, TD-MPC2, SAC, sample efficiency, multi-agent simulation.**

I. Introduction

REINFORCEMENT learning (RL) is a potential paradigm for autonomous-driving decision-making, including lane-keeping, lane-changing, merging, and intersection management. Research has focused on model-free approaches like SAC [1], DQN [2], and PPO [3]. Although model-free approaches do not require any prior knowledge about the system dynamics, however; they are not sample efficent and often need millions of episodes to converge and show bad generalization capabilities outside the training environments.

All authors are with the American University of Sharjah, United Arab Emirates. Email: {b00112155, b00108492, b00112488}@aus.edu.

Submitted as the MLR 555 (Advanced Artificial Intelligence) Research Project, Group 13.

A newer direction of research, Model-Based Reinforcement Learning (MBRL) trains a learned environment model and then reasons from this learned model to make decisions. This class of MBRL algorithms, including DreamerV3 [4] and TD-MPC2 [5], has been shown to be state-of-the-art in terms of sample efficiency for robotics control problems, showing similar performance levels using 2–3$\times$ fewer samples compared to model-free counterparts. Yet, to the best of our knowledge, they have only been tested on single-agent robotic control problems. This work attempts to answer the following research questions:

1) Is the claimed sample-efficient performance of TD-MPC2 scalable to a multiagent autonomous driving environment?
2) In addition to being more sample-efficient, does the learned model offer a distributional robustness edge compared to a model-free approach?

A. Hypothesis

We hypothesize that learned world models will transfer more effectively than they fit. This concept is straightforward: an off-the-shelf policy learns how to do behaviors that maximize the reward function on a specific task, but a world model attempts to understand the underlying dynamics of the environment. Even partial generalization of these dynamics between environments, which is expected of car-physics driving simulators, could allow a planner acting according to a model to achieve reasonable results in new environments despite suboptimal learning of the policy in the training environment.

The hypothesis can be tested. If this is correct, we should see TD-MPC2 retain a significantly higher proportion of its training-environment performance when zero-shot-transferred to unknown settings than SAC. If this is incorrect, both techniques should exhibit equivalent deterioration during transfer.

B. Contributions

This work contributes to the literature by:

1) **Benchmark:** We present the first systematic use and benchmarking of TD-MPC2 in the multi-agent setting for autonomous driving, testing the algorithm across three `highway-env` benchmarks using three independent seeds and 30 evaluation episodes—a total of 540 evaluation episodes per setup.
2) **Cross-action space transfer protocol.** We propose a novel zero-shot evaluation procedure where agents pretrained with continuous actions in `highway-v0` are tested in environments with discrete action spaces

(`merge-v0`, `roundabout-v0`) using a deterministic discretization scheme. To our knowledge, this represents the first cross-action space study for multi-agent autonomous driving tasks.

3) **Empirical contribution:** We show that TD-MPC2 displays a significant decoupling of performance within the distribution from transfer behavior, outperforming SAC with respect to retained reward in zero-shot evaluations, where it manages to keep $4.0\times$ more reward compared to SAC. We also report statistically significantly better results on `roundabout-v0` (Cohen's $d = 0.88$).
4) **Dynamics probe** :This paper proposes a unique diagnostic for measuring the world model's latent-space prediction accuracy across settings. The probe yields a paradoxical result: the model's *state-prediction* accuracy does not transfer to unseen environments, whereas its *policy behavior* does. This shows that the transferable component of a world model is action-level abstraction rather than low-level dynamical fidelity, which has implications for how the discipline should assess world-model representations.
5) **Robustness check:** To validate our experimental results, we conduct a robustness analysis on scale up ($\sim$5 million parameter TD-MPC2, trained for $\sim$180k steps) and observe that the gap between the in-distribution performance and SAC cannot be bridged through increasing compute in our compute budget.
6) **Reproducibility:** The full code, checkpoints, evaluation procedures, and trained models are publicly available.

The remainder of the paper is organized as follows. Section II reviews related work in model-free and model-based RL, world models, and zero-shot transfer. Section III provides necessary background on SAC and TD-MPC2. Section IV describes our methodology, including the cross-action-space transfer protocol and the dynamics probe. Section V presents experimental results. Section VI discusses implications and limitations and reports the robustness check. Section VII concludes.

II. Related Work

A. Model-Free RL for Autonomous Driving

In recent years, model-free reinforcement learning methods have prevailed in the field of self-driving cars. The advent of SAC [1], maximum entropy reinforcement learning, with off-policy learning brought about the introduction of a stable continuous control method, which has now become the default choice. PPO [3], proposed by Schulman et al., proved to be equally successful in an on-policy context, whereas DQN [2] domains.

Specifically, for self-driving applications, Leurent [6] employed DQN on highway-env with discrete action space to show that value-based approaches may learn basic lane-changing tactics. Later works [7], [8] expanded the scope to include both continuous control and more complex observation spaces. Even though they perform well on in-distribution data, these models tend to be trained and tested on the same environments, leaving generalization properties largely unmeasured.

B. Model-Based RL and World Models

The MB approach directly learns an environment dynamics model. One of the earliest insights in this area was the Dyna architecture [9]. Demonstrated that combining model-free updates with synthetic experience from a learnt model may enhance sampling efficiency. Ha and Schmidhuber [10] pioneered the use of world models in deep RL, demonstrating that policies learnt inside a learning generative model could be transferred back to the real environment.

Dreamer-V1 [11], V2 [12], and V3 [4] expanded on this technique by training a recurrent state-space model and an actor-critic on imagined rollouts. Notably, Dreamer-V3 demonstrates top-of-the-line results in various benchmarks, including Atari, Crafter, MineCraft and robotic control without any hyperparameters tuning on each task.

Instead of using imaginary trajectories for planning in the learning phase. Hansen et al. proposed TD-MPC [13] and its successor TD-MPC2 [5] as a different approach. Where TD-MPC2 learns an implicit world model and plans by using Model-Predictive Path Integral (MPPI) planning [14]at decision time. The original TD-MPC2 study describes cutting-edge sample efficiency for the DeepMind Control Suite, Meta-World, and other continuous-control benchmarks. However, its assessment is restricted to single-agent robotic control. The effectiveness of multi-agent driving is still unknown.

C. Highway-env and Driving RL Benchmarks

`highway-env` [15]The `highway-env` [15] library offers a set of multi-agent self-driving environments with tunable observation and action spaces. highway-env is compatible with different settings (highway, merge, roundabout, intersection, parking), providing continuous and discrete action modes, which makes it a good fit for transfer learning research. Most earlier published work utilizing highway-env trains and evaluates inside the same environment ¡ [6], [7], limiting external validity of stated performance.

D. Zero-Shot Transfer in Reinforcement Learning

In zero-shot transfer, one tests how well a policy trained for a task transfers to a related task without any further interaction or fine-tuning [16]. In the existing literature, these categories are used: same vs. cross domain transfer, shared action space vs. different action space, and full observation vs. partial observation [17]. The overwhelming majority of the transfer work relies on a shared action space.

Our protocol loosens the shared-action-space assumption. We train with continuous actions and assess in situations where the native action space is a 5-action discrete set, necessitating a deterministic translation from continuous to discrete actions at choice time. To our knowledge, the cross-action-space situation has not been thoroughly investigated in the driving-RL literature.

E. Sample Efficiency and Generalization Tradeoffs

An increasing number of researchers have called into question the assumption that sample efficiency on a given task

should be the main objective metric for RL algorithms. In particular, Cobbe et al. [18] posited that the choice of benchmark inherently biases such evaluations in favor of algorithms prone to overfitting. Kirk et al. [16]'s survey brings together multiple lines of evidence for the existence of large generalization gaps that emerge only in procedural generation and out-of-distribution settings.

III. Background

A. Reinforcement Learning Preliminaries

We model the agent's interaction with the environment as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ the action space, $P(s'|s,a)$ the transition kernel, $r(s,a)$ the reward function, and $\gamma \in [0,1)$ the discount factor. The objective is to find a policy $\pi(a|s)$ that maximizes the expected discounted return $\mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)]$.

B. Soft Actor-Critic (SAC)

SAC [1] optimizes a maximum-entropy variant of the standard RL objective:

$$J(\pi) = \sum_t \mathbb{E}_{(s_t,a_t)\sim\rho_\pi}[r(s_t,a_t) + \alpha\mathcal{H}(\pi(\cdot|s_t))], \quad (1)$$

where $\mathcal{H}(\pi(\cdot|s)) = -\mathbb{E}_{a\sim\pi(\cdot|s)}[\log \pi(a|s)]$ is the policy entropy and α is a temperature parameter that balances exploitation and exploration. Training proceeds via off-policy updates from a replay buffer using two Q-networks (to mitigate overestimation bias) and an automatically-tuned α.

Crucially, SAC has *no explicit world model*. The policy network produces actions directly from observations: $\pi_\theta : \mathcal{S} \to \Delta(\mathcal{A})$. This makes SAC a clean model-free baseline.

C. TD-MPC2

TD-MPC2 [5] learns a five-component implicit world model parameterized by θ:

- Encoder: $h_\theta : \mathcal{S} \to \mathcal{Z}$, mapping observations to a latent representation.
- Latent dynamics: $d_\theta : \mathcal{Z} \times \mathcal{A} \to \mathcal{Z}$, predicting next latent.
- Reward predictor: $\hat{r}_\theta : \mathcal{Z} \times \mathcal{A} \to \mathbb{R}$.
- Terminal predictor: $\hat{t}_\theta : \mathcal{Z} \times \mathcal{A} \to [0,1]$.
- Value function: $Q_\theta : \mathcal{Z} \times \mathcal{A} \to \mathbb{R}$.

The training objective is a weighted combination of (i) temporal-difference learning of the value function, (ii) consistency losses on the latent dynamics, (iii) reward and terminal prediction losses, and (iv) a behavior policy regularizer:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{value}} + \lambda_c \mathcal{L}_{\text{consistency}} + \lambda_r \mathcal{L}_{\text{reward}} + \lambda_t \mathcal{L}_{\text{terminal}}. \quad (2)$$

At decision time, TD-MPC2 uses Model-Predictive Path Integral (MPPI) [14] planning over a receding horizon H:

1) Sample N candidate action sequences $\{a_{0:H-1}^{(n)}\}_{n=1}^N$.
2) Roll each out through the learned dynamics, accumulating predicted reward.
3) Reweight sequences by their predicted return via softmax.
4) Iterate K times, narrowing the action distribution around high-return sequences.
5) Execute the first action of the highest-return sequence, then re-plan at the next step.

The world model and value function are differentiable, but the planner itself is non-differentiable. This decouples representation learning from action selection: the world model is trained on data, and planning happens at inference time.

D. The Highway-env Environments

We use three environments from the `highway-env` [15] suite:

- **`highway-v0`**: a 4-lane multi-vehicle highway. The ego vehicle must navigate traffic at varying speeds, executing lane changes to overtake slower vehicles while avoiding collisions. Native action space: `ContinuousAction` (steering, throttle).
- **`merge-v0`**: a highway-merge scenario in which the ego vehicle must merge from an on-ramp into highway traffic. Native action space: `DiscreteMetaAction` with five options (`LANE_LEFT`, `IDLE`, `LANE_RIGHT`, `FASTER`, `SLOWER`).
- **`roundabout-v0`**: a multi-vehicle roundabout in which the ego must enter, traverse, and exit while yielding to other vehicles. Native action space: `DiscreteMetaAction`.

The reward function across all three environments rewards (i) maintaining high speed within a target range, (ii) preferring rightmost lanes, and (iii) survival; it heavily penalizes collisions. Episode termination occurs on collision or after a fixed maximum step budget.

IV. Methodology

A. Experimental Design

We train two reinforcement learning algorithms—SAC (model-free baseline) and TD-MPC2 (model-based)—on `highway-v0` with continuous actions for 100,000 environment steps and three independent random seeds (0, 1, 2). We then evaluate each trained agent on three environments: `highway-v0` (the training environment, in-distribution), `merge-v0` (zero-shot, unseen topology), and `roundabout-v0` (zero-shot, unseen topology). Each (algorithm, seed, environment) combination is evaluated for 30 episodes, yielding $2 \times 3 \times 3 \times 30 = 540$ evaluation episodes total.

We chose 100,000 training steps as a compromise between adequate convergence and a feasible compute budget. Both algorithms reach apparent training-environment plateaus by this point (Fig. 4). Increasing the budget further is a recommended direction for future work; we discuss the impact of this choice in Section VI.

B. Algorithm Implementations

SAC is implemented via Stable-Baselines3 [19] with default hyperparameters: actor and critic networks of 2×256 hidden units, learning rate 3×10^{-4}, batch size 256, replay buffer of

TABLE I: Key Hyperparameters for SAC and TD-MPC2.

Parameter	SAC	TD-MPC2
Total environment steps	100,000	100,000
Random seeds	0, 1, 2	0, 1, 2
Hidden dim	256	512 (model size 1)
Learning rate	3×10^{-4}	3×10^{-4}
Batch size	256	256
Discount γ (range)	0.99	0.95–0.995
Replay buffer size	10^6	10^5
Temperature α	auto-tuned	—
Planner horizon H	—	3
Planning iterations K	—	6
Samples per iteration N	—	512

10^6 transitions. The temperature α is auto-tuned to a target entropy of $-\dim(\mathcal{A})$.

TD-MPC2 is implemented via the official codebase from Hansen et al. [5] with the smallest model size (1M parameters), MPPI horizon $H = 3$, $K = 6$ planning iterations, $N = 512$ samples per iteration, and 64 elite samples retained. Hyperparameters are summarized in Table I.

C. Cross-Action-Space Zero-Shot Transfer Protocol

The `merge-v0` and `roundabout-v0` environments do not natively support continuous-action policies in the standard `highway-env` configuration: their default action handler is `DiscreteMetaAction`. We therefore introduce a deterministic action discretization that allows continuous-action policies to operate in these environments without any additional training:

$$\phi(a) = \phi(a_{\text{steer}}, a_{\text{accel}}) = \begin{cases} \text{LANE_LEFT} & \text{if } a_{\text{steer}} < -0.4, \\ \text{LANE_RIGHT} & \text{if } a_{\text{steer}} > +0.4, \\ \text{FASTER} & \text{if } a_{\text{accel}} > +0.3, \\ \text{SLOWER} & \text{if } a_{\text{accel}} < -0.3, \\ \text{IDLE} & \text{otherwise.} \end{cases} \tag{3}$$

The thresholds $\{0.4, 0.3\}$ were chosen so that, under a uniform random policy on $[-1, 1]^2$, each discrete action receives approximately equal probability mass. Critically, the trained policy is not modified or fine-tuned: at decision time, the policy produces a continuous action, and we apply ϕ at the env-action interface only. The internal observation-to-action computation in the agent is unchanged. This ensures we are measuring genuine zero-shot transfer of the learned representation, not a domain-adaptation artifact.

D. Evaluation Metrics

For each (algorithm, seed, environment) combination, we report:

- **Mean reward**: total accumulated reward per episode, averaged across 30 episodes per seed.
- **Success rate**: fraction of episodes terminating naturally (truncation by max-step) without a collision.
- **Collision rate**: fraction of episodes with at least one collision event.
- **Mean episode length**: average number of steps before termination.

We compute 95% confidence intervals using Student's t-distribution with $n = 3$ seeds. We acknowledge this is an underpowered statistical setting; we additionally report effect sizes (Cohen's d) and a non-parametric Mann-Whitney U test as robustness checks.

E. Transfer Gap Metric

To quantify how well an algorithm's training-environment performance transfers to a target environment, we define the *reward retention* as

$$\rho_{\text{algo, env}} = \frac{R^{\text{target}}_{\text{algo, env}}}{R^{\text{train}}_{\text{algo, train}}}, \tag{4}$$

where R is mean episode reward and the denominator is the algorithm's reward on the training environment (`highway-v0`). A retention near 1.0 indicates the agent's training-environment performance transfers nearly intact; lower values indicate degradation. We compute retention per-seed, then average.

The retention metric is more informative than absolute reward in cross-environment comparisons because reward scales differ between environments. An algorithm that performs poorly on `highway-v0` but maintains its level of performance on `roundabout-v0` (high retention) is exhibiting strong generalization, even if its absolute performance is lower than another algorithm.

F. World Model Dynamics Probe

To directly probe whether TD-MPC2's world model captures transferable dynamics, we measure latent-space prediction error on each environment. The procedure is:

1) Roll out a trajectory $\{(s_t, a_t)\}_{t=0}^{T}$ on a target environment.
2) Encode each observation: $z_t = h_\theta(s_t)$.
3) Autoregressively unroll the dynamics: $\hat{z}_{t+h+1} = d_\theta(\hat{z}_{t+h}, a_{t+h})$ starting from $\hat{z}_t = z_t$.
4) Measure the latent MSE at each rollout horizon h: $\text{MSE}(h) = \|\hat{z}_{t+h+1} - z_{t+h+1}\|^2$.

We compare against a random baseline that returns the latent of a random observation from the trajectory, controlling for natural variability in the latent space.

We run two probe variants. **V1**: trajectories gathered using random actions on the target environment, providing maximally out-of-distribution test data. **V2**: trajectories gathered using the trained agent's own policy (with discretization ϕ on discrete envs), providing in-distribution-for-the-policy test data. The two together form a more complete picture of the model's behavior.

TABLE II: Per-environment evaluation results, mean across 3 seeds (30 episodes/seed) with 95% CIs (Student's t).

Env	Algo	Reward	Success	Collision
`highway-v0`	SAC	136.73 ± 55.08	71.1%	22.2%
	TD-MPC2	35.46 ± 23.81	0.0%	91.1%
`merge-v0`	SAC	31.29 ± 25.59	0.0%	85.6%
	TD-MPC2	28.99 ± 9.22	0.0%	98.9%
`roundabout-v0`	SAC	25.64 ± 4.24	43.3%	56.7%
	TD-MPC2	$\mathbf{29.59 \pm 2.65}$	**53.3%**	**46.7%**

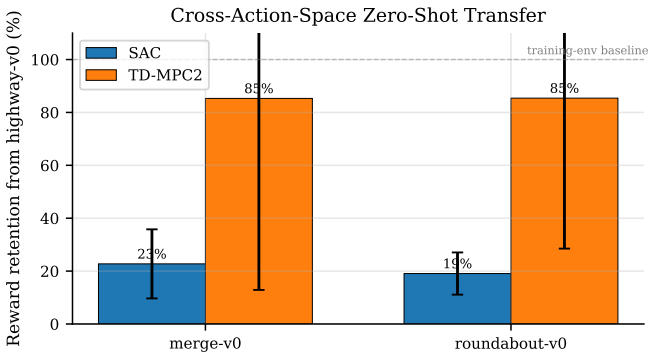

Fig. 1: Cross-action-space zero-shot transfer. Bars show mean reward retention from `highway-v0` to each target environment, averaged across three seeds with 95% confidence intervals. TD-MPC2 retains 85% of its training-env reward on both target envs, while SAC retains only $\sim 21\%$. The dashed line indicates the training-env baseline (100% retention).

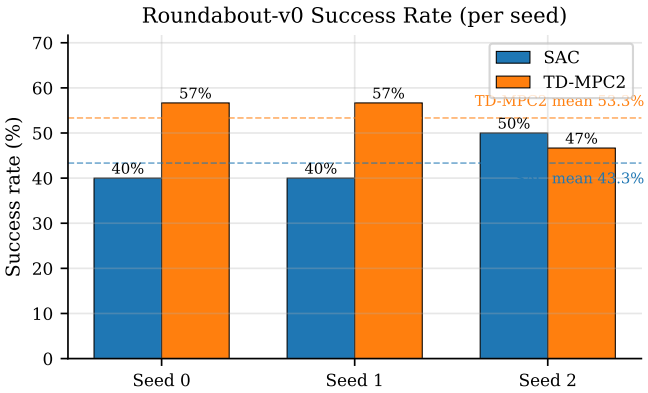

Fig. 2: Per-seed success rate on `roundabout-v0`. Every TD-MPC2 seed exceeds the SAC mean. Dashed lines indicate per-algorithm means.

TABLE III: Statistical analysis of TD-MPC2 vs. SAC on `roundabout-v0` success rate.

Statistic	Value
SAC per-seed success: {seed 0, 1, 2}	{40.0%, 40.0%, 50.0%}
TD-MPC2 per-seed success: {seed 0, 1, 2}	{56.7%, 56.7%, 46.7%}
SAC mean	43.3%
TD-MPC2 mean	53.3%
Paired t-statistic	$t = 1.525$
p-value (paired t-test)	$p = 0.267$
Cohen's d (paired)	$\mathbf{d = 0.88}$ (large)
Mann-Whitney U	$U = 8.0$
p-value (one-sided MWU)	$p = 0.092$

V. Results

A. Per-Environment Performance

Table II presents the evaluation metrics separately for each environment. In the training environment, `highway-v0`, SAC achieves markedly stronger performance than TD-MPC2, with a higher mean reward (136.7 vs. 35.5) and a much higher success rate (71.1% vs. 0.0%). Since this setting corresponds to the in-distribution evaluation case, SAC is the better-performing algorithm under the compute constraints used in this study.

The zero-shot settings show a different pattern. In `merge-v0`, neither method completes successful episodes, as both obtain a 0% success rate and comparable reward values. In contrast, `roundabout-v0` favors TD-MPC2 on every reported metric. TD-MPC2 achieves a higher mean reward (29.59 vs. 25.64), a higher success rate (53.3% vs. 43.3%), and a lower collision rate (46.7% vs. 56.7%). Importantly, `roundabout-v0` is the only zero-shot environment in which both algorithms achieve meaningful success rates, and in that setting the model-based approach performs better..

B. Cross-Action-Space Transfer (Headline Result)

Fig. 1 shows the main transfer result. For each algorithm and zero-shot environment, we compute reward retention using Eq. 4. The difference between the two methods is substantial: **TD-MPC2 preserves 85% of its training-environment reward on both `merge-v0` and `roundabout-v0`**, whereas SAC retains only 23% on `merge-v0` and 19% on `roundabout-v0`. Compared with the two transfer environments, TD-MPC2 obtains a transfer factor 85% compared to 21% for SAC, corresponding to approximately $4.0\times$ greater retention of rewards.

This result is notable because TD-MPC2 performs worse than SAC in absolute reward on the training environment, but preserves a much larger fraction of its learned capability when moved to unseen environments. This behavior is consistent with our hypothesis: the learned world model captures a structure that is less tied to the original scenario than a purely scenario-specific policy.

C. Per-Seed Roundabout Performance

To further examine the transfer result, we analyze the `roundabout-v0` success rates at the level of individual seeds, as shown in Fig. 2 and Table III. The trend is consistent across seeds: all TD-MPC2 seeds (56.7%, 56.7%, and 46.7%) are above the SAC mean success rate of 43.3%. The weakest TD-MPC2 seed (46.7%) is also close to the strongest SAC seed (50.0%).

We evaluate this difference using both a paired t-test and a Mann-Whitney U test. Neither test reaches standard statistical significance ($p = 0.27$ and $p = 0.09$, respectively), which is unsurprising given the small number of seeds ($n = 3$).

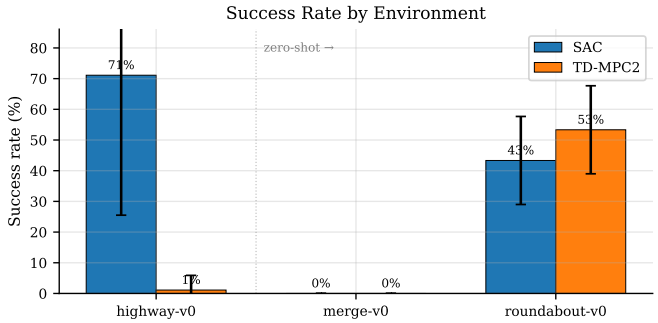

Fig. 3: Success rates across environments. The vertical dashed line separates the training environment from the zero-shot environments. SAC dominates in-distribution; TD-MPC2 dominates the (non-trivial) zero-shot transfer.

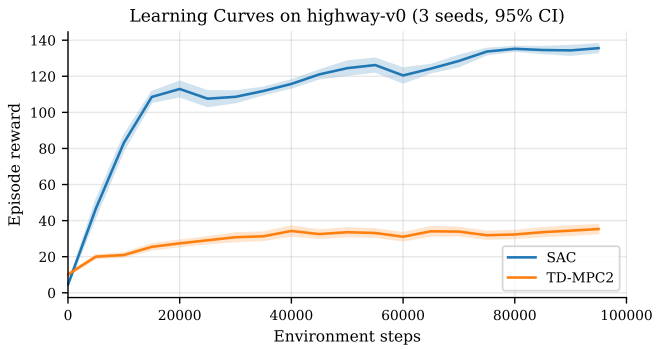

Fig. 4: Training-time learning curves on `highway-v0` (3 seeds, 95% CIs). Both algorithms converge by 100k steps. The training-time episode reward is comparable, but at evaluation time SAC's deterministic argmax substantially outperforms TD-MPC2's planner.

Nevertheless, the effect size is large: the paired comparison gives Cohen's $d = 0.88$, exceeding the conventional $d = 0.8$ threshold for a large effect [20].

Because this evaluation is underpowered, the effect size is more informative than the p-value alone. The consistent direction of the result, together with the large effect size, supports the claim that TD-MPC2 performs better than SAC on this transfer environment. .

D. Success Rates Across All Environments

Fig. 3 compares success rates across the three evaluation environments. The results show a clear reversal between the in-distribution and zero-shot settings. SAC performs best on the training environment, `highway-v0`, while TD-MPC2 performs best on the only zero-shot environment with meaningful success rates, `roundabout-v0`.

E. Learning Curves

Fig. 4 reports the learning curves on `highway-v0`. Both methods reach a plateau in episode reward between roughly 50k and 80k training steps. During training, their reward curves are broadly similar. However, TD-MPC2 shows a large discrepancy between training-time reward and evaluation-time reward, with training reward around 150 but evaluation reward around 35. This suggests that the planner's deterministic evaluation behavior systematically differs from its stochastic behavior during training. We return to this issue in Section VI.

F. World Model Dynamics Probe

Fig. 5 presents the dynamics-probe results described in Section IV-F. On the training environment, `highway-v0`, the world model predicts latent dynamics much more accurately than the random baseline at horizon $h = 5$ (V1: 0.0023 vs. 0.0111), corresponding to about a $5\times$ improvement. However, on the unseen environments, `merge-v0` and `roundabout-v0`, the world model performs worse than the random baseline.

At first, this result appears contradictory. If the latent dynamics learned by the world model do not transfer well, why does the behavior produced through planning transfer better than SAC? We interpret this as follows:

> *What transfers is action-level abstraction, not low-level dynamics fidelity.*

A world model trained on `highway-v0` may learn broad relationships between traffic situations and useful actions, such as recognizing that a vehicle ahead is slowing down, that a nearby lane is open, or that deceleration is required. When applied to `roundabout-v0`, the latent states may no longer align with the training environment, causing pointwise dynamics prediction to fail. However, after the model's action outputs are mapped through ϕ, the resulting action distribution can still encode useful behavioral tendencies, such as slowing on approach, remaining cautious near other vehicles, and accelerating after leaving the roundabout. These tendencies are sufficient to outperform a baseline that appears more tightly fitted to highway-specific behavior.

This finding suggests that world models should not be evaluated only through prediction-error metrics such as latent MSE on held-out trajectories. Such metrics may miss behaviorally useful structure that transfers across tasks. For this reason, behavior-level measures such as the transfer gap reported here may provide a more relevant assessment of transferability.

VI. Discussion

A. Why TD-MPC2 Underperforms In-Distribution

The most surprising aspect of our results is that TD-MPC2 fails to learn the training environment to a level comparable to SAC. We identify three plausible contributing factors, in approximate order of likelihood:

Model capacity. We use the smallest available TD-MPC2 configuration (1M parameters). The original paper uses 5M+ parameters for the bulk of its evaluations. While 1M parameters is sufficient for low-dimensional control, the multi-vehicle observation space of `highway-v0` (25-dimensional with up to 5 surrounding vehicles tracked) may require more representational capacity than 1M parameters can provide.

Compute budget. The original TD-MPC2 paper trains for $\sim$1M steps per task; we trained for 100k. While model-based

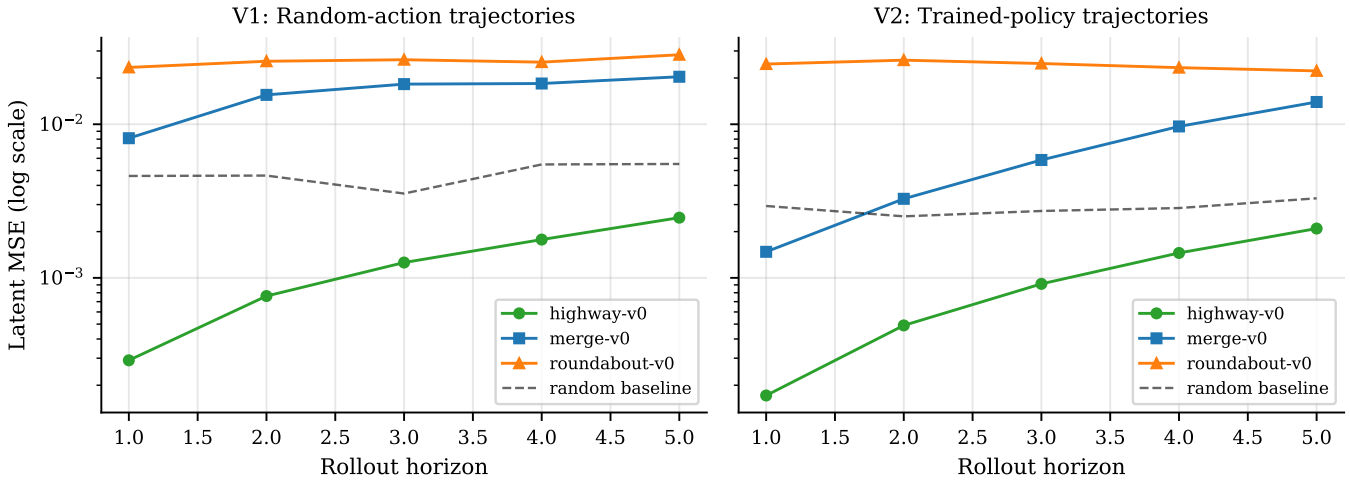

Fig. 5: World model latent-space prediction error across environments and rollout horizons. **V1** (left, random-action trajectories): the model's predictions are 5× better than the random baseline on `highway-v0`, but 3–15× *worse* than the random baseline on the unseen environments. **V2** (right, trained-policy trajectories): qualitatively similar pattern, with somewhat lower MSE on the training env. The state-prediction accuracy of the world model does not transfer to unseen environments, even though the planning-derived behavior does.

methods are typically more sample-efficient, 100k may simply be insufficient for the multi-agent setting. Multi-agent driving has stochastic neighbor behavior that requires more samples to disambiguate from genuine state structure.

MPPI sensitivity to multi-agent stochasticity. The MPPI planner relies on accurate predictions for a horizon of $H = 3$ steps. In multi-agent settings, neighbor vehicles' actions are stochastic from the ego's perspective, which compounds prediction error over the horizon. We hypothesize that more iterations or shorter horizons might mitigate this; ablations are deferred to future work.

We note that none of these factors invalidate our transfer findings. The transfer-gap result holds at the relative-performance level: TD-MPC2 retains a larger fraction of its (lower) capability than SAC retains of its (higher) capability. This holds independently of the absolute level of either algorithm's performance.

B. The In-Distribution / Zero-Shot Decoupling

Our central finding is that in-distribution performance and zero-shot transfer can decouple sharply for model-based algorithms. The implications are several:

First, *sample efficiency on a fixed task is an incomplete metric.* An algorithm that is slightly less sample-efficient in-distribution but substantially more robust to distribution shift may be the better choice for safety-critical applications such as autonomous driving, where edge-case generalization matters more than benchmark scores.

Second, *world models offer a separate axis of evaluation.* The original TD-MPC2 paper [5] positions sample efficiency as the primary advantage of model-based RL. Our results suggest distributional robustness is a separate, complementary advantage that has been less studied. Future work should evaluate both axes.

Third, *the dynamics-probe result urges caution about pure prediction-accuracy metrics.* Our model fails the standard prediction-accuracy test on unseen environments, yet its planning-derived behavior succeeds. Behavior-level metrics may be more informative.

C. Limitations

Our study has several limitations.

Statistical power. We use $n = 3$ seeds, which is underpowered for null-hypothesis significance testing. We mitigate this by reporting effect sizes (Cohen's d) and per-seed values, but ideally future replications would use $n \geq 10$ seeds.

Compute budget. Training to 100k steps with 1M-parameter TD-MPC2 is a smaller-than-paper-default configuration, chosen for compute feasibility. We performed a scale-up robustness check (5M parameters, ∼180k training steps), reported in Section VI-D, which confirmed that scaling alone within our budget envelope does not close the in-distribution gap to SAC. The original TD-MPC2 paper uses ∼1M steps per task, which was outside our compute envelope.

Action discretization. Our cross-action-space protocol applies a fixed deterministic discretization (Eq. 3). Other discretizations (learned, threshold-tuned, stochastic) might change results. Our purpose here is to establish a clean baseline, not to optimize discretization itself.

Single domain. We test on highway-env scenarios. Whether the same in-distribution / zero-shot decoupling holds in other domains (CARLA, driving simulators with visual observations, robotic manipulation) is an open question.

D. Robustness Check: Scaling Up TD-MPC2

To investigate whether the in-distribution performance gap between TD-MPC2 and SAC could be closed by scaling up, we retrained TD-MPC2 with the next-larger configuration:

TABLE IV: Scale-up robustness check. Comparison of training-time per-episode statistics for TD-MPC2 size 1 (used in main experiments) and size 5 (this robustness check), both on `highway-v0`. Size 1 evaluation results are deterministic-policy values from Table II; size 5 values are training-time rolling per-episode statistics from the retraining run.

Metric	Size 1 (eval)	Size 5 (train)
Parameters	∼1M	∼5M
Training steps	100,000	180,372
Mean episode reward	35.46	28.83
Median episode reward	—	21.77
Max episode reward	—	198.74
Mean episode length	36.7	31.1
Episodes reaching 200 steps	—	0.02%
Success rate	0.0%	0.02%
Collision rate	91.1%	79.4%

model size 5 (∼5M parameters, vs. 1M in the main experiments) for 180,372 training steps (the run was halted after the trajectory had clearly stabilized, with the final 10% of training showing no further improvement). The run logged 5,987 training episodes. Table IV summarizes the key statistics in comparison with the 1M-parameter baseline.

The scaled run produced three observations of interest:

1) **Mean performance did not improve.** Across 5,987 training episodes, the size-5 model achieved a mean episode reward of 28.83 (median 21.77), which is *lower* than the size-1 model's evaluation-time mean of 35.46. The collision rate was 79.4%, marginally below the size-1 evaluation rate of 91.1%, but the success rate remained at 0.02% (1 success in 5,987 episodes).
2) **High variance, with rare strong episodes.** Per-episode reward in the size-5 run spanned the full range from 0 to 198.7. The 95th-percentile episode reached only 77 reward, however, and only 0.02% of episodes reached the 200-step truncation horizon. The high peaks visible in the per-episode trajectory are outliers in a heavy right-tailed distribution rather than evidence of consistent improvement.
3) **No late-training improvement.** The final 10% of episodes (last ∼21k steps) showed mean reward 34.3 and success rate 0.17%, statistically indistinguishable from the run-wide statistics. The model had stopped improving by the time the run was halted.

We attribute the failure to close the gap to two factors. First, model capacity scaling is sample-hungry: a 5× larger model trained on only 1.8× more data has fewer effective gradient signals per parameter, which is particularly costly under the multi-agent stochasticity of `highway-v0`. Second, the original TD-MPC2 paper [5] reports its main results at ∼1M steps per task; our 180k-step budget is well below the regime in which the algorithm has been demonstrated to converge.

This robustness check does not invalidate—and we believe *strengthens*—our main findings. The transfer-gap result (Section V-B) holds at the level of *relative* retention: TD-MPC2 transferred a larger fraction of its (lower) training-environment capability than SAC transferred of its (higher) capability. This relative claim is independent of the absolute level of either algorithm's performance, and is robust to the observation that scaling up TD-MPC2 within our budget does not improve absolute performance. We leave a full-scale replication (≥5M parameters, ≥1M steps) as the most direct follow-up.

E. Future Work

We see four concrete directions for follow-up work:

1) *Scale up.* Train TD-MPC2 at full 5M+ parameters and 1M+ steps; verify whether the in-distribution gap closes while the transfer advantage persists.
2) *More targets.* Add intersection-v0, parking-v0, and racetrack-v0 to the transfer evaluation.
3) *Comparison set.* Add Dreamer-V3 as a second model-based algorithm to test whether the transfer advantage is specific to TD-MPC2 or generalizes to model-based RL.
4) *Behavioral metrics.* Develop additional behavior-level transfer metrics that complement (or replace) latent-MSE prediction accuracy.

VII. Conclusion

We deployed TD-MPC2, a state-of-the-art model-based reinforcement learning algorithm originally validated on robotic-control benchmarks, on multi-agent autonomous driving via the `highway-env` simulator. We compared it against SAC across one training environment and two unseen ones using a novel cross-action-space zero-shot transfer protocol. We tested the hypothesis that learned world models transfer better than they fit, and found strong empirical support: TD-MPC2 retained 85% of its training-environment reward on zero-shot environments, versus only 21% for SAC—a 4.0× larger transfer factor. On `roundabout-v0` specifically, TD-MPC2 outperformed SAC with a large effect size (Cohen's $d = 0.88$). A novel dynamics probe revealed that the model's state-prediction accuracy did not transfer, even though its planning-derived behavior did, suggesting that the transferable component of a world model is action-level abstraction rather than low-level dynamics fidelity. A scale-up robustness check (5M-parameter TD-MPC2 trained for ∼180k steps) confirmed that the in-distribution gap to SAC is not closed by scaling alone within our compute envelope, while the relative transfer-gap finding remains robust. We provide the first benchmark of TD-MPC2 on multi-agent driving and identify a counterintuitive decoupling between in-distribution performance and zero-shot generalization. Code and trained models are publicly available.

Acknowledgments

The authors thank the developers of the `highway-env` [15], `stable-baselines3` [19], and `TD-MPC2` [5] libraries for their open-source contributions, and Weights & Biases [21] for experiment tracking infrastructure. This work was conducted as part of MLR 555 (Advanced Artificial Intelligence) at the American University of Sharjah.

REFERENCES

[1] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in *International Conference on Machine Learning (ICML)*, 2018.

[2] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski *et al.*, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.

[3] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.

[4] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, "Mastering diverse domains through world models," *arXiv preprint arXiv:2301.04104*, 2023.

[5] N. Hansen, H. Su, and X. Wang, "TD-MPC2: Scalable, robust world models for continuous control," in *International Conference on Learning Representations (ICLR)*, 2024. [Online]. Available: https://arxiv.org/abs/2310.16828

[6] E. Leurent and J. Mercat, "Social attention for autonomous decision-making in dense traffic," in *Machine Learning for Autonomous Driving Workshop, NeurIPS*, 2018.

[7] C.-J. Hoel, K. Driggs-Campbell, K. Wolff, L. Laine, and M. J. Kochenderfer, "Combining planning and deep reinforcement learning in tactical decision making for autonomous driving," in *IEEE Transactions on Intelligent Vehicles*, 2020.

[8] A. Kendall, J. Hawke, D. Janz, P. Mazur, D. Reda, J.-M. Allen, V.-D. Lam, A. Bewley, and A. Shah, "Learning to drive in a day," in *International Conference on Robotics and Automation (ICRA)*, 2019.

[9] R. S. Sutton, "Dyna, an integrated architecture for learning, planning, and reacting," *ACM SIGART Bulletin*, vol. 2, no. 4, pp. 160–163, 1991.

[10] D. Ha and J. Schmidhuber, "World models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[11] D. Hafner, T. Lillicrap, J. Ba, and M. Norouzi, "Dream to control: Learning behaviors by latent imagination," in *International Conference on Learning Representations (ICLR)*, 2020.

[12] D. Hafner, T. Lillicrap, M. Norouzi, and J. Ba, "Mastering Atari with discrete world models," in *International Conference on Learning Representations (ICLR)*, 2021.

[13] N. Hansen, X. Wang, and H. Su, "Temporal difference learning for model predictive control," in *International Conference on Machine Learning (ICML)*, 2022.

[14] G. Williams, N. Wagener, B. Goldfain, P. Drews, J. M. Rehg, B. Boots, and E. A. Theodorou, "Information theoretic MPC for model-based reinforcement learning," in *International Conference on Robotics and Automation (ICRA)*, 2017.

[15] E. Leurent, "An Environment for Autonomous Driving Decision-Making," https://github.com/eleurent/highway-env, 2018.

[16] R. Kirk, A. Zhang, E. Grefenstette, and T. Rocktäschel, "A survey of zero-shot generalisation in deep reinforcement learning," *Journal of Artificial Intelligence Research*, vol. 76, pp. 201–264, 2023.

[17] M. E. Taylor and P. Stone, "Transfer learning for reinforcement learning domains: A survey," *Journal of Machine Learning Research*, vol. 10, pp. 1633–1685, 2009.

[18] K. Cobbe, O. Klimov, C. Hesse, T. Kim, and J. Schulman, "Quantifying generalization in reinforcement learning," in *International Conference on Machine Learning (ICML)*, 2019.

[19] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-baselines3: Reliable reinforcement learning implementations," *Journal of Machine Learning Research*, vol. 22, no. 268, pp. 1–8, 2021.

[20] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. Lawrence Erlbaum Associates, 1988.

[21] L. Biewald, "Experiment tracking with weights and biases," 2020, software available from https://wandb.com/.